

"Visual Design Language & Components"

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1. Design Philosophy

IdeaDNA's design follows a 'Dark DNA' aesthetic — futuristic, scientific, and developer-friendly. The visual language draws from biotechnology (DNA helix, molecular structures) and cyberpunk (neon accents, dark backgrounds). Every element serves a purpose: information density is high, animations guide attention, and colors communicate meaning.

- Clarity Over Decoration: Every pixel serves a functional purpose
- Progressive Disclosure: Show essentials first, details on demand
- Motion with Meaning: Animations guide user attention, not distract
- Accessibility First: WCAG 2.1 AA compliance, keyboard navigation, screen readers
- Developer Comfort: Dark theme default, monospace for code, high contrast

2. Color System

2.1 Primary Palette

Token	Hex	Usage
--color-primary	#1a1a2e	Main background, headers, cards
--color-secondary	#16213e	Secondary background, panels, sidebars
--color-accent	#e94560	CTA buttons, highlights, active states, DNA strands
--color-success	#00d9ff	Success states, high scores, positive indicators
--color-warning	#ffc107	Medium scores, warnings, pending states
--color-danger	#ff4757	Errors, low scores, critical alerts
--color-background	#0f0f23	Page background, void areas
--color-surface	#1e1e3f	Elevated surfaces, modals, dropdowns
--color-text-primary	#ffffff	Headings, primary text
--color-text-secondary	#a0a0b0	Body text, descriptions
--color-text-muted	#6c6c80	Captions, timestamps, placeholders
--color-border	#2a2a4a	Dividers, card borders, input borders

2.2 Score Color Coding

Range	Color	Meaning
90-100	#00d9ff (Cyan)	Exceptional — Top tier idea
80-89	#28a745 (Green)	Strong — Excellent candidate
70-79	#ffc107 (Yellow)	Good — Solid with room for improvement
60-69	#fd7e14 (Orange)	Average — Needs work
50-59	#ff4757 (Red)	Weak — Consider alternatives
<50	#dc3545 (Dark Red)	Reconsider — Major issues

3. Typography System

Element	Font	Size	Weight	Line Height	Usage
Hero	Inter	28px	800	1.2	Landing page headline
H1	Inter	18px	700	1.3	Page titles, section headers
H2	Inter	14px	700	1.4	Card titles, feature names
H3	Inter	12px	700	1.4	Sub-sections, labels
Body	Inter	10px	400	1.5	Descriptions, content
Caption	Inter	10px	400	1.4	Timestamps, metadata
Code	JetBrains Mono	9px	400	1.6	Code snippets, tech stacks
Button	Inter	10px	600	1	CTA, actions

4. Spacing & Layout System

4.1 Spacing Scale (4px base unit)

- 0px (0) — None
- 4px (1) — Tight (icon gaps, inline spacing)
- 8px (2) — Small (button padding, card internal)
- 12px (3) — Compact (form field gaps)
- 16px (4) — Medium (section padding, card margins)
- 24px (6) — Large (component gaps, section dividers)
- 32px (8) — XLarge (page sections, hero spacing)
- 48px (12) — Huge (major section breaks)
- 64px (16) — Massive (page vertical rhythm)

4.2 Layout Grid

12-column grid system. Max container width: 1280px. Gutters: 24px. Breakpoints: Mobile 320px (1 col), Tablet 768px (2-4 col), Desktop 1024px+ (3-6 col).

5. Component Library

Button

Variants: Primary (accent bg), Secondary (border), Ghost (text only), Danger (red). Sizes: sm (32px), md (40px), lg (48px). States: default, hover, active, disabled, loading.

Card

Idea Card: Rounded 12px, border 1px, shadow on hover. Contains: title, description, score badges, tags, select button. Evolution Card: Same + mutation tag, parent indicator.

Input

Textarea: Auto-resize, character counter, validation states. Dropdown: Custom styled, searchable, grouped options. Toggle: Switch for boolean settings.

Modal

Centered overlay, backdrop blur, close on Escape/click outside. Sizes: sm (400px), md (600px), lg (800px), xl (full). Animations: fade + scale.

Loading States

DNA Helix Spinner: Rotating double helix for main loading. Skeleton: Shimmer effect for content placeholders. Progress Bar: Step indicator for multi-step flows.

Toast Notification

Positions: top-right, bottom-center. Types: success (cyan), error (red), warning (yellow), info (blue). Auto-dismiss: 5 seconds. Action button support.

Radar Chart

D3.js component. 4 axes, animated fill, hover tooltips. Colors match score ranges. Legend below.

Evolution Tree

React Flow custom nodes. Root = gray, evolved = color-coded by score. Edges animated with mutation labels. Pan, zoom, fit-view controls.

6. Animation Guidelines

All animations use Framer Motion with spring physics. Duration: 0.3s default, 0.5s for page transitions. Easing: spring (stiffness: 300, damping: 30) for UI, ease-in-out for fades.

Page Transitions

Fade + slide up (20px). Duration: 0.4s. Exit: fade out.

Card Hover

Lift (translateY -4px) + shadow increase. Duration: 0.2s.

Score Counter

Count up from 0 to final value. Duration: 1s. Easing: easeOut.

Radar Chart Fill

Gradual fill from center. Duration: 1.5s. Staggered per axis.

Tree Node Expand

Scale from 0.8 to 1 + fade in. Duration: 0.3s. Stagger: 0.1s per node.

Loading Helix

Continuous rotation (360deg / 2s). Pulse glow effect.

Toast Enter/Exit

Slide from right + fade. Duration: 0.3s. Exit: slide up + fade.

— End of DESIGN —